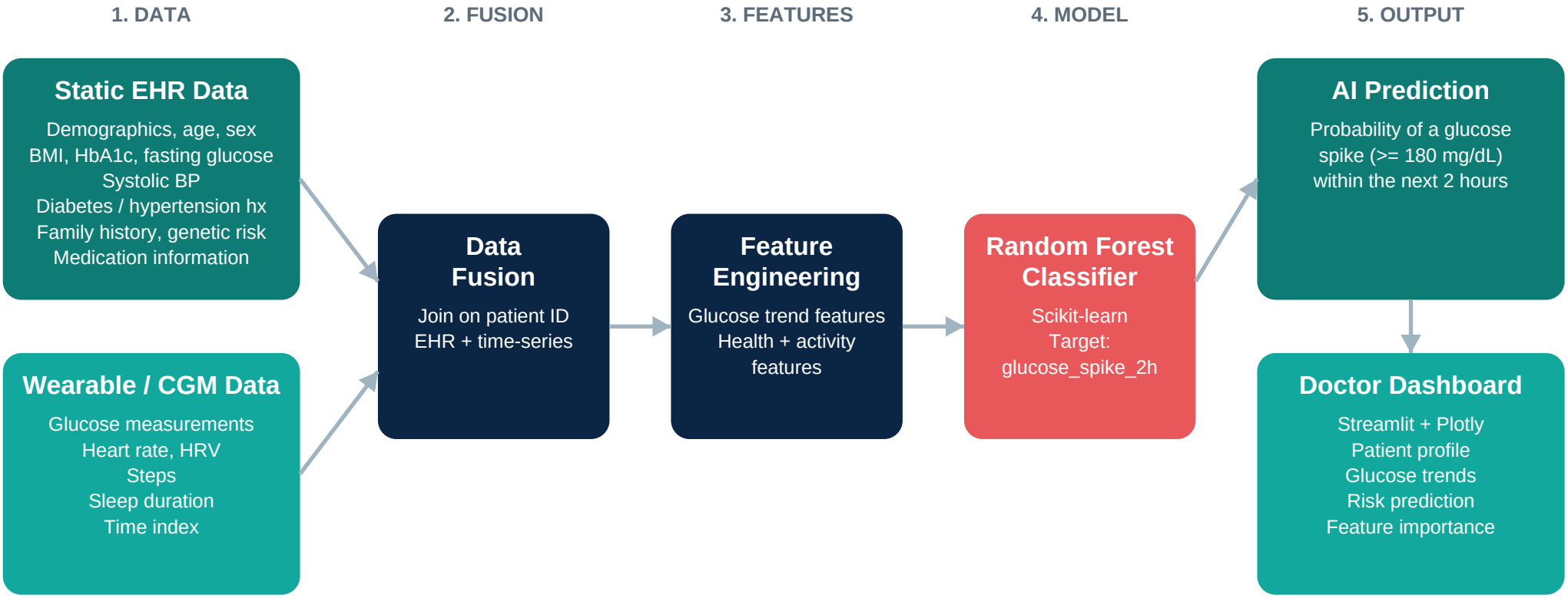
System Architecture: Digital Twin for Glucose Spike Prediction

Team Codexxx | Chandigarh University | Static EHR + wearable/CGM streams fused to predict a spike within 2 hours



How each stage maps to the repository

Data generation	`data/generate_ehr.py, data/generate_wearable.py -> ehr.csv, wearable.csv`
Fusion + features	`src/prepare_data.py -> data/fused_dataset.csv`
Training	`models/train_fused_model.py -> models/glucose_spike_model.pkl (saved with Joblib)`
Dashboard	`dashboard/app.py (streamlit run dashboard/app.py)`

All data is synthetic and generated locally; no real patient data is used (in line with DPDP Act / HIPAA constraints). Educational prototype, not a clinically validated system.

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